

Install guide — Field Box BT

App v0.8.0 · box firmware 0.3.1-bt

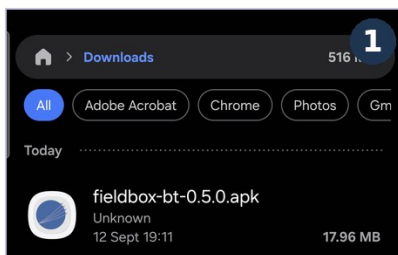
1. What you need

- An **Android 10 or newer** phone, with GPS and Bluetooth.
- The install file `fieldbox-bt-<version>.apk` (for example `fieldbox-bt-0.8.0.apk`).
- A **Field Box** with firmware **0.3.1-bt** (older boxes on 0.2.0-bt still work, but without continuous background logging — see §5).

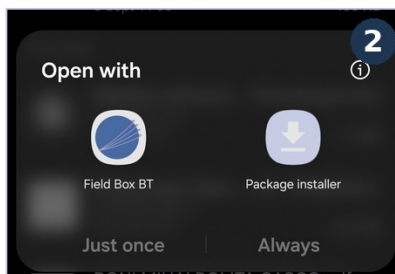
2. Installing the APK on the phone

The app is not on the Play Store; it is installed from an **APK** file sent by the coordinator. The steps below are from a Samsung phone on Android 14–16; on other phones the wording may differ slightly, the order is the same.

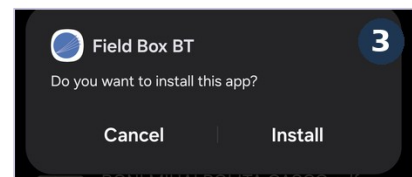
1. Get the file `fieldbox-bt-<version>.apk` onto the phone — via WhatsApp, a USB cable or Google Drive.
2. **Received on WhatsApp:** tap the file right in the chat (the first tap downloads it, the second opens it) — no need to go to Downloads; continue with step 3. **Received another way:** open **My Files** → **Downloads** and tap the file.
3. If an **"Open with"** window appears, choose **Package installer** (not Field Box BT — that is the app already installed, not the installer).
4. At **"Do you want to install this app?"** tap **Install**. The first time Android may ask to allow installs from that source (WhatsApp or My Files): a settings screen with **"Allow from this source"** appears — switch it on, go back and tap the file again.



My Files → Downloads: tap
fieldbox-bt-0.5.0.apk



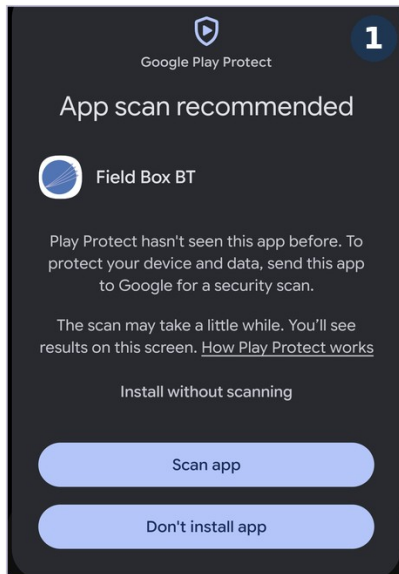
"Open with": choose Package
installer



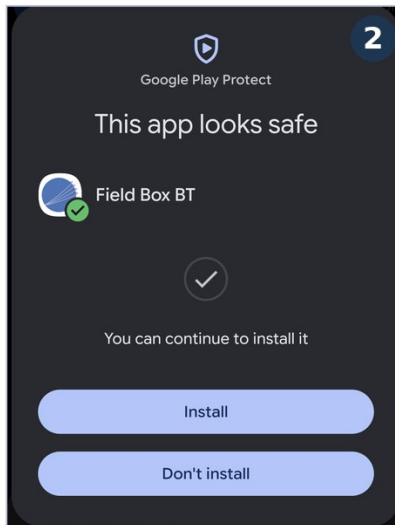
"Do you want to install this app?"
→ Install

Steps 2–4

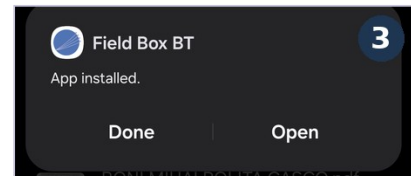
5. **Google Play Protect** — **"App scan recommended"** appears: the app is not from the Play Store, so Google does not know it. Tap **Scan app**; the scan takes a few seconds. Without internet, tap **More details** → **Install without scanning**.
6. At **"This app looks safe"** tap **Install**.
7. At **"App installed"** tap **Open**.



Google Play Protect: tap Scan app (or "Install without scanning" with no internet)



"This app looks safe" → Install



"App installed" → Open

Steps 5–7

The app appears on the phone as **Field Box BT**. A newer version is installed the same way, **over** the old one — the profile, files and bundles on the phone stay. The installed version is shown on the **Boxes** tab, under the connection status (App v0.8.0 · firmware v...).

3. First launch

On first open the app asks, in turn:

1. **Profile** — the pupil's/teacher's name and class/school (optional). They appear in the report and in file names. Tap **Save**, or **Skip** (the session is then identified by the box code only). Can be changed any time with **edit** on the Boxes tab.
2. **Camera** — "Allow Field Box BT to take pictures..." → **While using the app** (for the station photos).
3. **Location** — choose **Precise** (not Approximate!) and **While using the app**. Without "Precise" the route cannot be recorded.
4. **Notifications** → **Allow** (the permanent notification shows the link state and has the **Read now** button).

Profile 1

Name

Mihai

Group / school (optional)

The name appears in the report and in file names. Without a name, the session is identified only by the box's ID.

Skip Save

Profile: name and class (or Skip)
→ Save

2

Allow **Field Box BT** to take pictures and record video?

While using the app

Only this time

Don't allow

Camera → While using the app

3

Allow **Field Box BT** to access this device's location?

Precise Approximate

While using the app

Only this time

Don't allow

Location → Precise + While using the app

4

Allow **Field Box BT** to send you notifications?

Allow

Don't allow

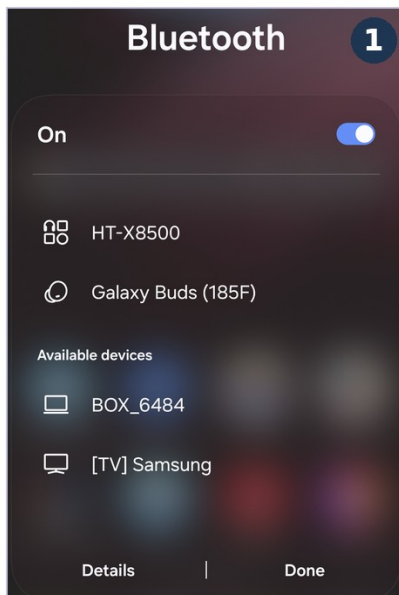
Notifications → Allow

First launch: profile and permissions

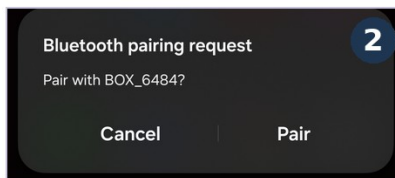
Bluetooth pairing (once, per phone and box)

The box must be **paired** with the phone before the app can see it.

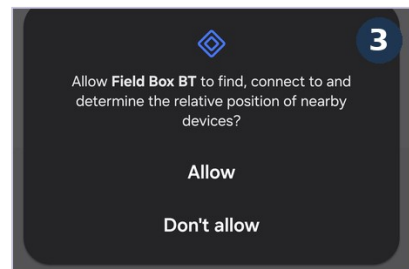
1. Power the box (the LED blinks). On the phone: **Settings → Bluetooth** (or long-press the Bluetooth icon in the quick settings).
2. Under **Available devices** the box appears as **BOX_xxxx** (the code on its label) — tap it.
3. At "**Bluetooth pairing request — Pair with BOX_xxxx?**" tap **Pair**. No PIN is asked. The box moves to "Paired devices"; the pairing is kept.
4. In the app, **Boxes → Connect to a box**. The first time Android asks for the **Nearby devices** permission → **Allow**.
5. In the **Choose a box** window tap **BOX_xxxx**. Within a few seconds the strip at the top turns green: **BOX_xxxx · link good**, and the card shows the box's firmware and clock (synchronised automatically).



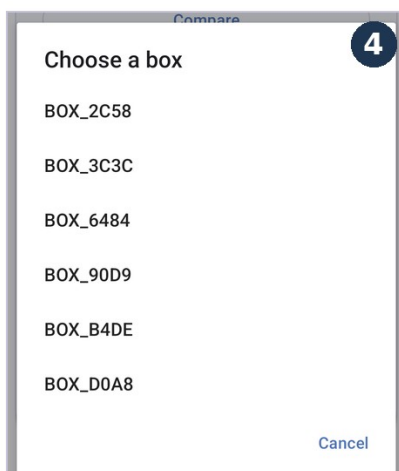
Settings → Bluetooth: tap BOX_xxxx under Available devices



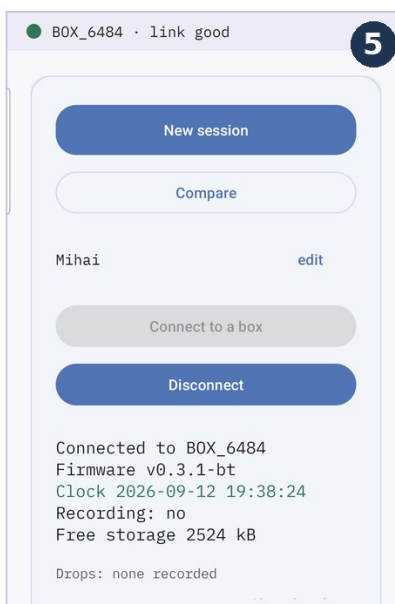
"Bluetooth pairing request" → Pair (once per phone and box)



In the app, Connect to a box → Nearby devices → Allow



Choose a box: tap BOX_xxxx



Connected: green dot, "link good", firmware and clock

Bluetooth pairing and the first connection

4. Quick check

Before heading out to the field, verify:

- The **Live** tab shows sensor values (not blank) once connected.
- The **Point** tab, held outdoors for 1–2 minutes, reaches a GPS fix (the pill turns green).
- The **Map** tab shows map tiles (not just grey), if the area has had signal at some point.

5. Updating the box (firmware) — for the technician only

This step is not for teachers — it is for whoever maintains the boxes.

- Firmware file: ``fieldbox-fw-<version>.bin``.
- Flashed at address ``0x0``, over USB, with **esptool** or the **Flash Download Tool**; on Windows this needs the **CH340** driver for the serial port.
- **The first boot after a firmware flash reformats the box's storage.** Get the files off the box first (**Files** tab: **Download** on each row, or **PDF report**, which fetches the day's log by itself) — anything still on it is erased by the update.
- The box's name (``BOX_xxxx``) comes from the board's Bluetooth MAC address and does not change with a firmware update.
- A box on **0.2.0-bt** still works with the 0.8.0 app, but without continuous background logging (the log only runs while a recording is active). Mixed fleets (some boxes updated, some not) are fine.

6. Uninstalling / data

- **Uninstalling the app** does not delete the files in **Documents/FieldBox** and **DCIM/FieldBox** — they stay on the phone.
- However, sessions imported for comparison, in **Documents/FieldBox/compare/**, are lost — on reinstalling, the pupils' ZIPs must be sent again.
- **To free up space:** from the **Files** tab, download and then delete old files from the box; on the phone, delete past days' files from **Documents/FieldBox** once they have been copied to a laptop.